

1. Business Idea with Its Roadmap

Business Idea

An **Online Learning & Skill Development Platform** focused on practical IT skills (programming, data science, AI tools) for students and fresh graduates.

Roadmap

- **Idea Generation:** Identify skill gaps between academic education and industry demand.
 - **Market Research:** Analyze target users, competitors, pricing, and demand.
 - **Planning:** Define features (courses, quizzes, certificates), business model, and cost.
 - **Development:** Build MVP (Minimum Viable Product).
 - **Testing:** Pilot with limited users and collect feedback.
 - **Launch:** Public release with marketing campaigns.
 - **Scaling:** Add advanced courses, partnerships, and international reach.
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2. Entrepreneurship and Business Development

Entrepreneurship Challenges

Local Level

- Limited funding access
- Infrastructure and internet issues
- Lack of mentorship
- Regulatory complexity

Global Level

- Intense competition
- Cultural and legal differences
- Currency and taxation issues
- Rapid technological change

Concerns to Highlight in Developing a Business Idea

- Market demand
- Financial feasibility
- Legal and ethical issues

- Scalability
- Risk management

Business Process Development

- Input (resources, capital, labor)
- Process (operations, development, quality control)
- Output (product or service delivery)
- Feedback (customer response and improvement)

Marketing and Communication (Initial Stage)

- Social media promotion
- Content marketing (blogs, videos)
- Email marketing
- Influencer collaboration
- Free trials or demos

Deployment Process

- Infrastructure setup
- Product launch
- Customer onboarding
- Monitoring performance
- Continuous improvement

3. Intellectual Property and Patent

Intellectual Property (IP) in Innovation

IP protects **ideas, inventions, designs, and creative works**, ensuring innovators get recognition and financial benefits.

Rules for Expressing Intellectual Property Rights

- Proper documentation
- Legal registration
- Confidentiality agreements
- Licensing agreements

Intellectual Property Policy

- Defines ownership
- Encourages innovation
- Prevents misuse
- Supports commercialization

Patent

a) What is a Patent?

A **patent** is a legal right granted by the government that gives an inventor exclusive rights to make, use, or sell an invention for a fixed period (usually 20 years).

b) How to Format a Product as a Patent?

- Title of invention
- Background and problem statement
- Detailed description
- Claims (core legal protection)
- Drawings or flowcharts
- Abstract

c) IT Industry Patent Examples

- Data compression algorithms
- Encryption techniques
- Database management systems
- AI model architectures

4. Innovation, Strategy, and Business Models

Innovation and Technology Adoption Theory

This theory explains how users adopt technology:

- Innovators
- Early adopters
- Early majority
- Late majority

- Laggards

For software commercialization, focus first on **early adopters**.

Competitive Strategy and Market Structure

- Cost leadership (low price)
- Differentiation (unique features)
- Focus strategy (niche market)

Market structures include monopoly, oligopoly, monopolistic competition, and perfect competition.

Pricing, Servicing, and Monetization

- Subscription model
- Freemium model
- Pay-per-use
- Advertising revenue
- Licensing

Business Model Development (IT Entrepreneurship)

- Value proposition
- Customer segments
- Revenue streams
- Cost structure
- Key partners
- Distribution channels

5. Engineering Ethics and Professionalism

Engineering Ethics (Definition)

Engineering ethics are **moral principles** that guide engineers to act responsibly, safely, and honestly.

Why Engineering Ethics Matters

- Protects public safety
- Ensures trust
- Prevents misuse of technology
- Promotes sustainability

Moral Issues in the Workplace

- Data privacy
- Workplace discrimination
- Corruption
- Safety negligence
- Conflict of interest

Consensus and Controversy

- **Consensus:** Shared agreement on ethical standards
- **Controversy:** Ethical disagreements due to values, culture, or 利益 conflicts

Profession and Professionalism

A **profession** requires specialized knowledge.

Professionalism means responsibility, integrity, and accountability.

Professional Roles of an Engineer

- Problem solver
- Innovator
- Ethical decision-maker
- Team leader
- Socially responsible citizen

6. Morality, Ethics, and Organizational Behavior

Define Morality

Morality refers to personal beliefs about **right and wrong**.

Ethics vs Morality

- Morality: Personal and cultural
- Ethics: Systematic rules and professional standards

Ethics as a Subset of Morality

Ethics are formalized principles derived from broader moral values.

Ethics in Organizational Environment

- **Educational:** Fair evaluation, no plagiarism

- **Official:** Transparency, accountability
- **Banking:** Confidentiality, honesty, anti-fraud practices

Leadership Ethics (Examples)

- Fair decision-making
- Transparency in communication
- Accountability for actions

Employee Duties, Responsibilities, and Rights

Duties: Obey rules, perform tasks

Responsibilities: Maintain integrity, teamwork

Rights: Fair wages, safe environment, respect